

Disaster Recovery Plan

Condensed Study Notes

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DISASTER RECOVERY PLAN

A Disaster Recovery Plan (DRP) is a documented process to help an organization quickly resume operations after an unplanned event. It outlines procedures to restore IT infrastructure and data. The goal is to minimize downtime and data loss.

Key aspects include:

- **Recovery Time Objective (RTO):** The maximum acceptable delay before critical systems must be back online.
- **Recovery Point Objective (RPO):** The maximum acceptable amount of data loss, measured in time (e.g., losing data from the last hour).

Think of it like a fire drill for your business's technology. You hope you never need it, but if disaster strikes, a DRP ensures you know exactly what steps to take to get things running again as fast and smoothly as possible.

How disaster recovery works

An effective Disaster Recovery Plan (DRP) addresses three key elements for recovery:

- **Preventive:** Focuses on making systems secure and reliable to stop disasters before they happen. This includes backing up data and monitoring for errors or compliance issues.
- **Detective:** Aims to identify unwanted events in real-time, signaling when a response is needed for rapid recovery.
- **Corrective:** Involves planning for DR scenarios, setting up backup operations to minimize impact, and executing procedures to quickly restore data and systems when a disaster occurs.

What is considered a disaster

A disaster is an event that disrupts normal operations and necessitates recovery procedures. Types of disasters include:

- **Natural disasters:** Such as earthquakes, floods, tornados, hurricanes, and wildfires.
- **Technological hazards:** Including power outages, pipeline explosions, and transportation accidents.
- **Pandemics and epidemics:** Widespread outbreaks of disease.
- **Cyber attacks:** Malicious actions like malware, Distributed Denial of Service (DDoS) attacks, and ransomware.
- **Machine and hardware failure:** Malfunctions of equipment.
- **Intentional, human-caused threats:** Such as terrorist or biochemical attacks.

What is a Disaster Recovery Plan

A Disaster Recovery Plan (DRP) is a structured, documented approach for an organization to quickly resume operations after an unplanned incident.

It's a critical component of a Business Continuity Plan (BCP).

DRP focuses on restoring IT infrastructure functionality and access after a disaster, ensuring IT systems supporting critical functions are operational quickly.

Organizations with well-tested DR strategies minimize disruption impact, achieve faster recovery, and resume core operations rapidly.

Disaster recovery planning is vital for all businesses, especially cloud-based ones, as disruptive events can occur without warning.

7 Chapters of an IT Disaster Recovery Plan

An IT Disaster Recovery Plan (DRP) is typically structured into seven key chapters. These chapters guide the creation of a comprehensive strategy for IT system restoration after a disaster.

While the specific content of each chapter isn't detailed here, their collective purpose is to provide a roadmap for recovery, ensuring that IT operations can resume as quickly and effectively as possible following an incident.

Here is the typical structure of a DR plan

A structured IT Disaster Recovery Plan (DRP) is typically organized into seven chapters:

1. **Goals:** Defines organizational objectives for disaster recovery, including the Recovery Time Objective (RTO) – the maximum acceptable downtime for critical systems – and the Recovery Point Objective (RPO) – the maximum acceptable data loss.
2. **Personnel:** Identifies individuals responsible for executing the DR plan.
3. **IT Inventory:** A comprehensive list of hardware and software assets, detailing their criticality and ownership status (leased, owned, or service).
4. **Backup Procedures:** Outlines the exact methods, locations (devices and folders), and recovery processes for all data resources.
5. **Disaster Recovery Procedures:** Details emergency responses to limit damage, includes last-minute backup protocols, and specifies mitigation and eradication steps, particularly for cybersecurity threats.
6. **Disaster Recovery Sites:** Describes the location and setup of alternative data centers (hot sites) in remote areas, equipped with replicated or frequently backed-up critical systems, ready for immediate takeover.
7. **Restoration:** Encompasses the procedures for recovering from complete system loss back to full operational status.

Why is a DR plan important

A Disaster Recovery Plan (DRP) is crucial for several reasons:

- **Minimizes Disruptions:** It helps reduce interruptions to normal operations.
- **Economic Protection:** It aims to lessen the financial impact of an event.
- **Rapid Restoration:** It ensures a smooth and quick return of services.
- **Prepared Operations:** It establishes backup methods for operating in advance.
- **Limits Damage:** It works to contain the scope of disruption and damage.
- **Personnel Training:** It provides training for staff on emergency procedures.

Planning a disaster recovery strategy

To plan effectively for disaster recovery, focus on these key metrics:

- **Recovery Point Objective (RPO):** The maximum acceptable age of data that can be lost. This metric informs how frequently backups should be performed.
- **Recovery Time Objective (RTO):** The maximum acceptable duration for systems and applications to be unavailable. This varies by application; some might tolerate an hour of downtime, while others need to be restored within minutes.

Types of Disaster Recovery Sites

Selecting a secondary site for data storage is crucial for preventing data loss. DR software uses this site to restore critical data to primary servers if they fail.

There are three main types of disaster recovery sites:

1. **Cold Computing Sites:** The simplest type. These sites provide only power, networking, and cooling. They lack essential hardware like servers and storage, requiring backups and additional hardware to be transported and installed before use, significantly impeding workflow.
2. **Warm Computing Sites:** Include all elements of a cold site plus storage hardware (tape/disk drives), servers, and switches. They are partially ready but still need data to be transported for recovery.
3. **Hot Computing Sites:** A fully functional backup site with important data already mirrored. This is the ideal, but often challenging, type of DR site.

Elements of Disaster Recovery Plan

A comprehensive Disaster Recovery Plan (DRP) is built upon several key elements:

1. **Disaster Recovery Team:** This team is responsible for creating, implementing, and maintaining the DRP. It requires clearly defined member roles, contact information, and instructions on who to contact during an emergency. All employees must be aware of the DRP and their roles.

2. Risk Identification and Assessment: The DR team identifies and evaluates potential risks, including natural disasters, man-made incidents, and technological failures. This assessment helps determine the necessary recovery strategies and resources needed to restore operations within acceptable timeframes.

3. Critical Application, Document, and Resource Identification: Organizations must pinpoint which business processes, applications, and resources are vital for immediate operational survival (e.g., payroll, cash flow generation). The focus is on short-term survivability.

4. Backup and Off-site Storage Procedures: This involves detailing what needs to be backed up (critical applications, equipment, documents like financial statements, employee lists, inventory, customer/vendor data, operational supplies, and the DRP itself), who performs the backup, how it's done, where backups are stored, and their frequency. Critical items must be stored off-site.

5. Testing and Maintenance: DRPs are living documents that require continuous refinement. Regular testing evaluates the plan's effectiveness and appropriateness. The plan must be updated to reflect changes in business processes, technology, and emerging disaster risks.